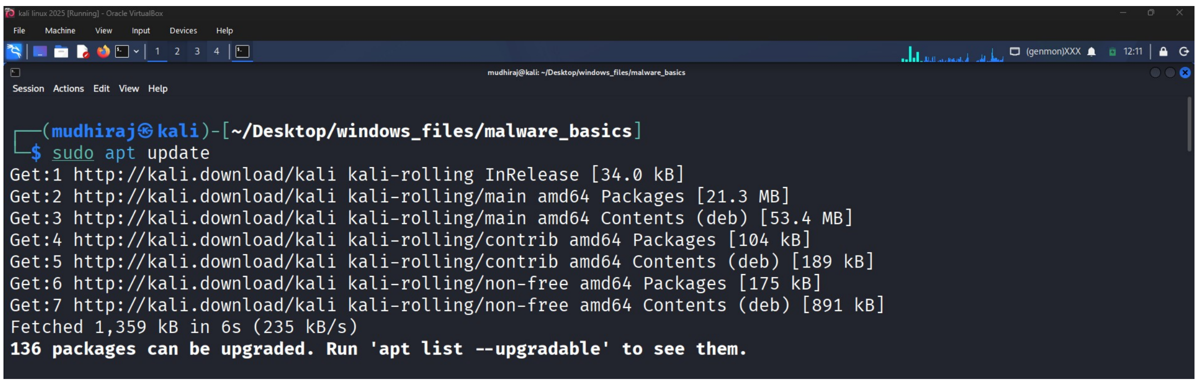
Task 4 – System Hardening

Objective:

Perform basic system hardening by applying updates, configuring firewall rules using iptables, and disabling an unused service.

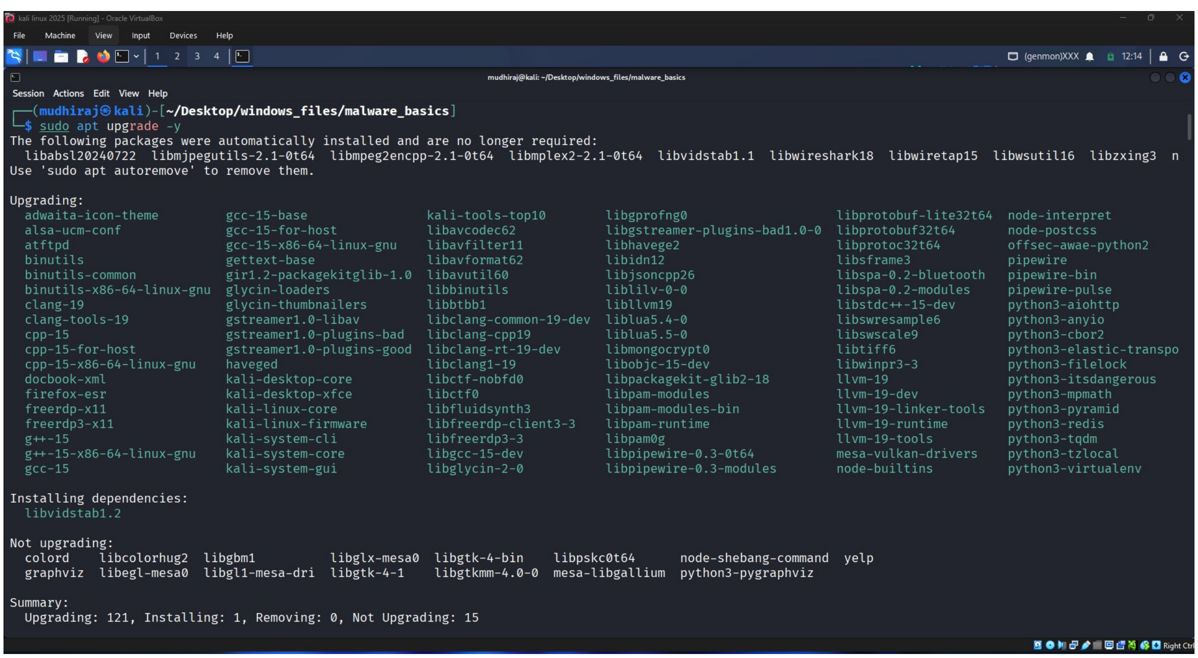
1 – System Update:

Executed `sudo apt update` to refresh package repositories.



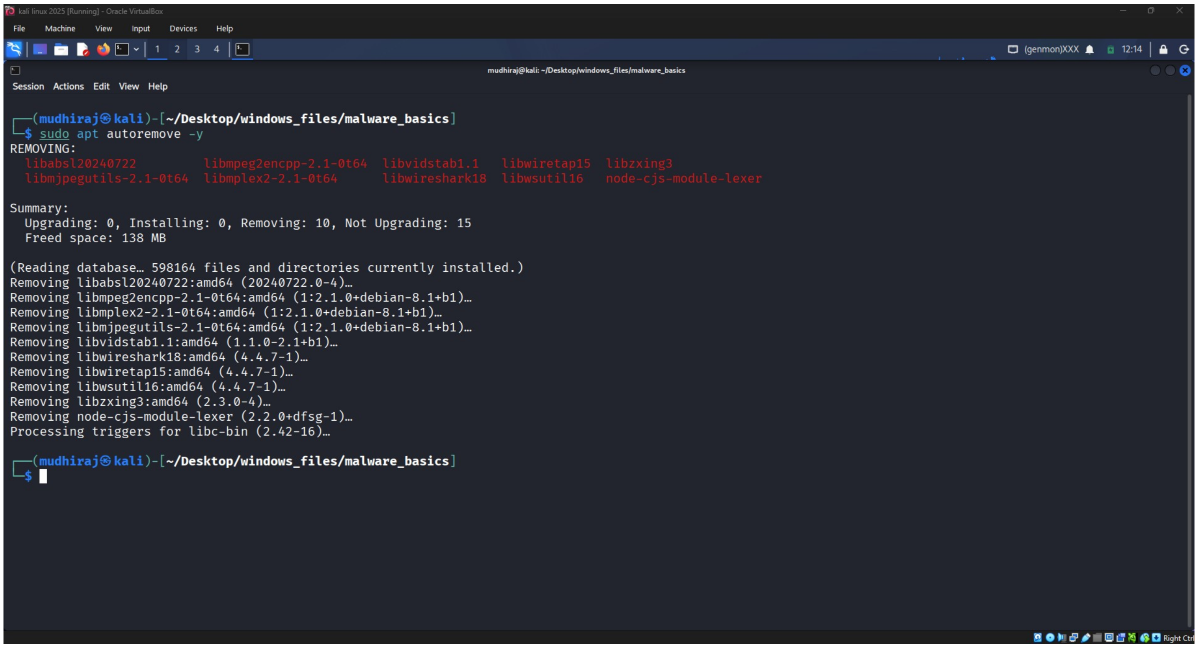
2 – Package Upgrade:

Executed `sudo apt upgrade -y` to install available security updates.



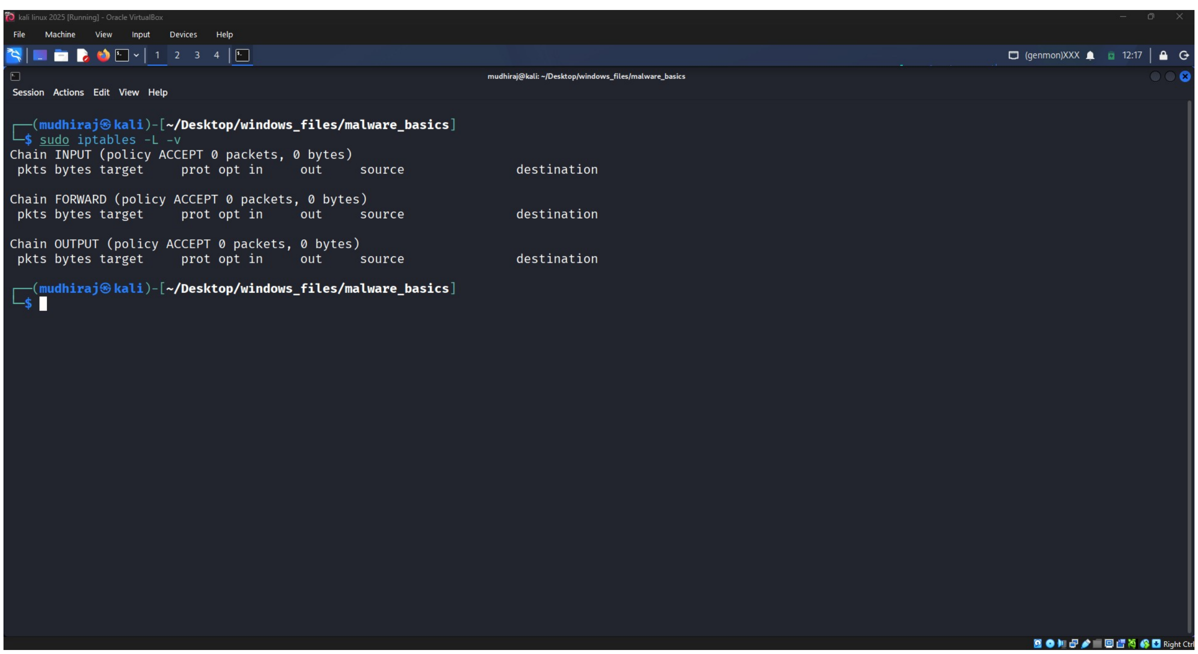
3 – Remove Unused Packages:

Executed `sudo apt autoremove -y` to remove unnecessary packages.



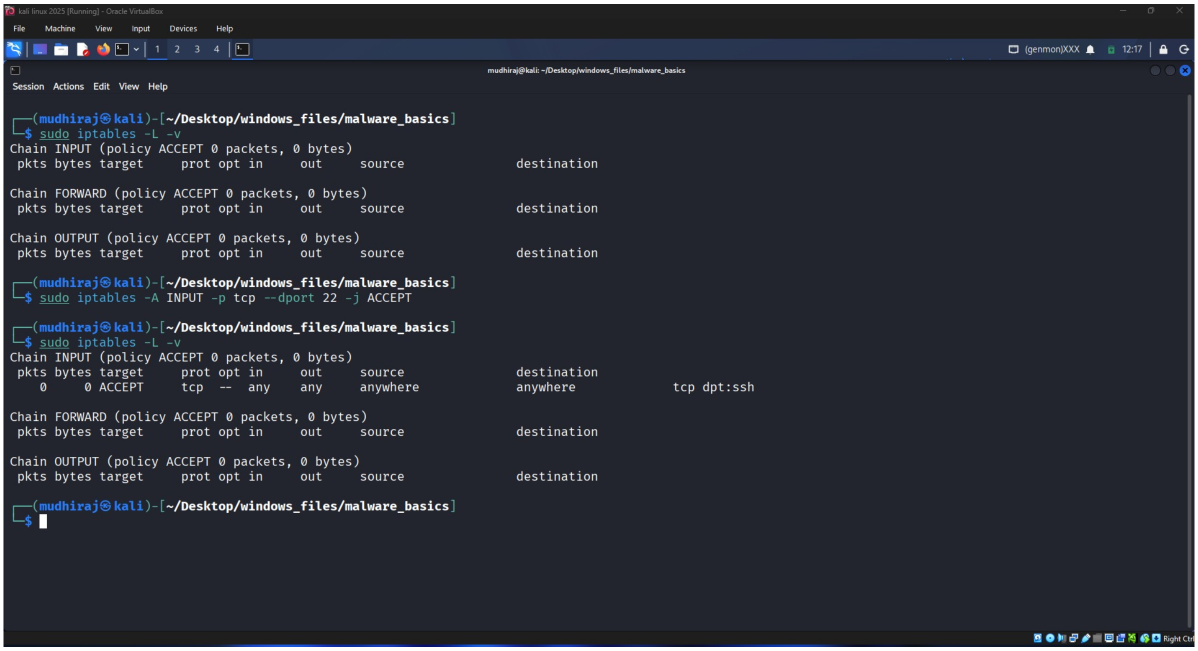
4 – Initial Firewall Rules:

Displayed the current iptables rules using `sudo iptables -L -v`.



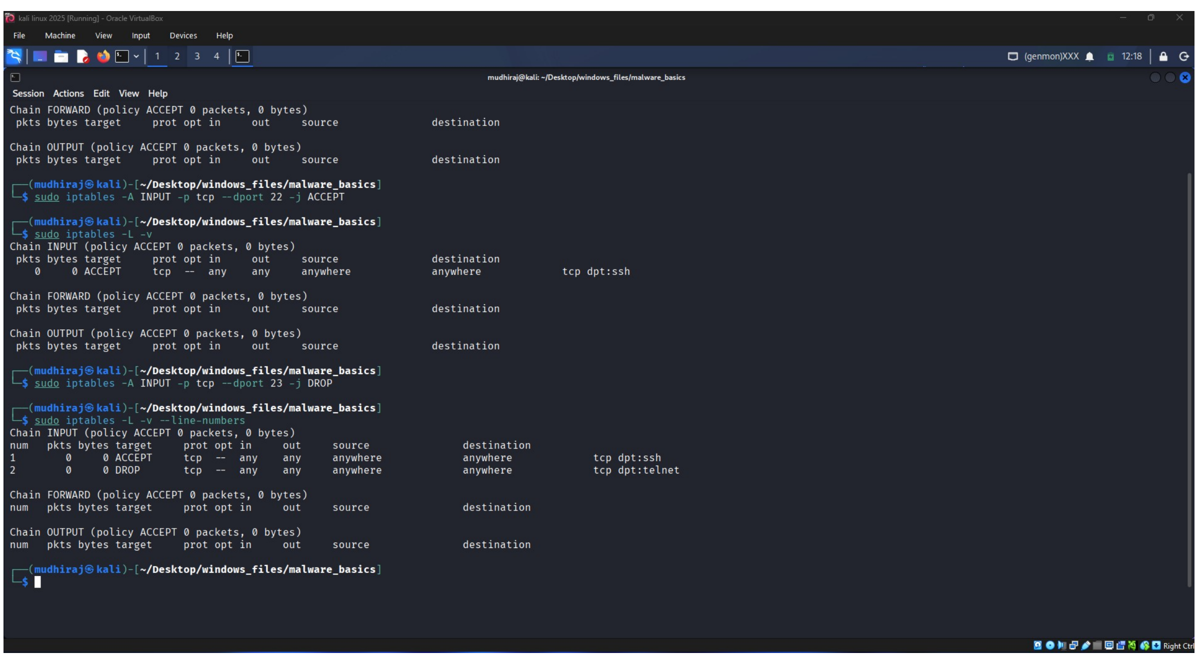
5 – Allow SSH:

Added a rule allowing inbound SSH traffic (TCP port 22).



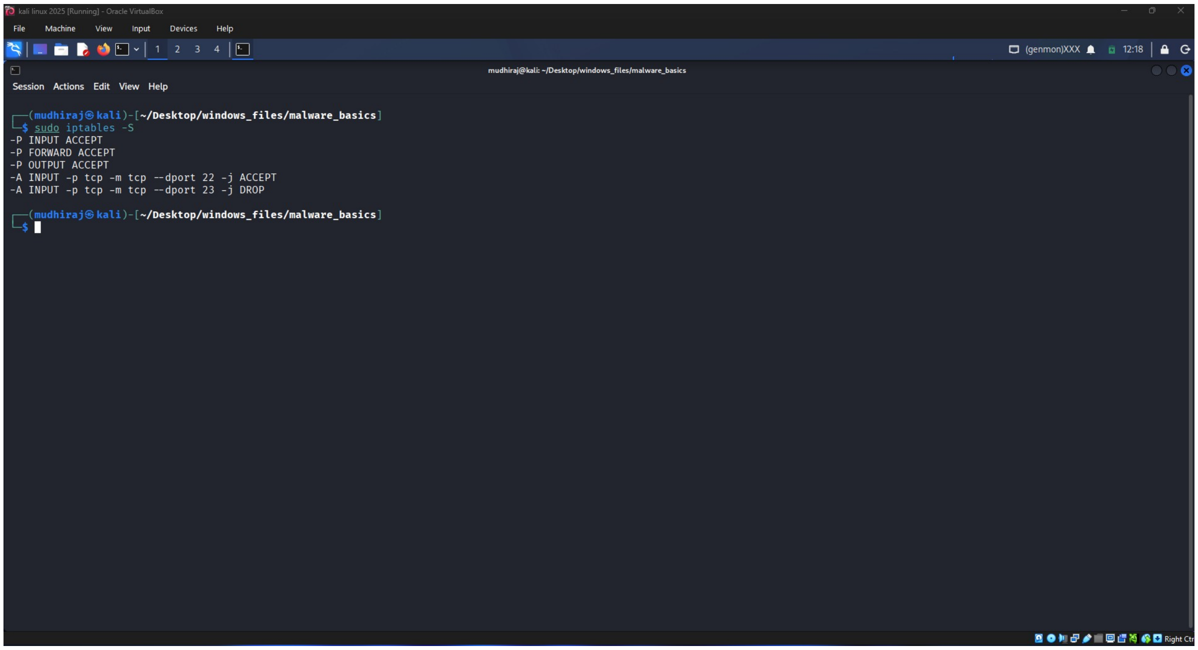
6 – Block Telnet:

Added a DROP rule for Telnet (TCP port 23) and verified with line numbers.



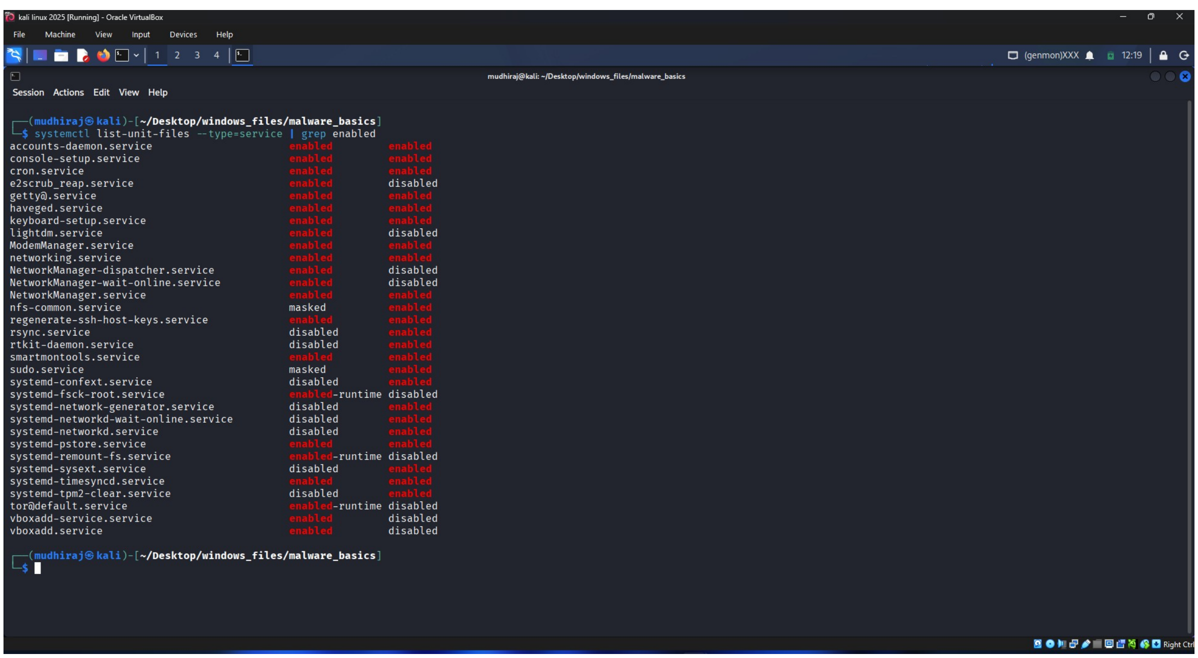
7 – Firewall Rule Summary:

Verified firewall rules using `sudo iptables -S`.



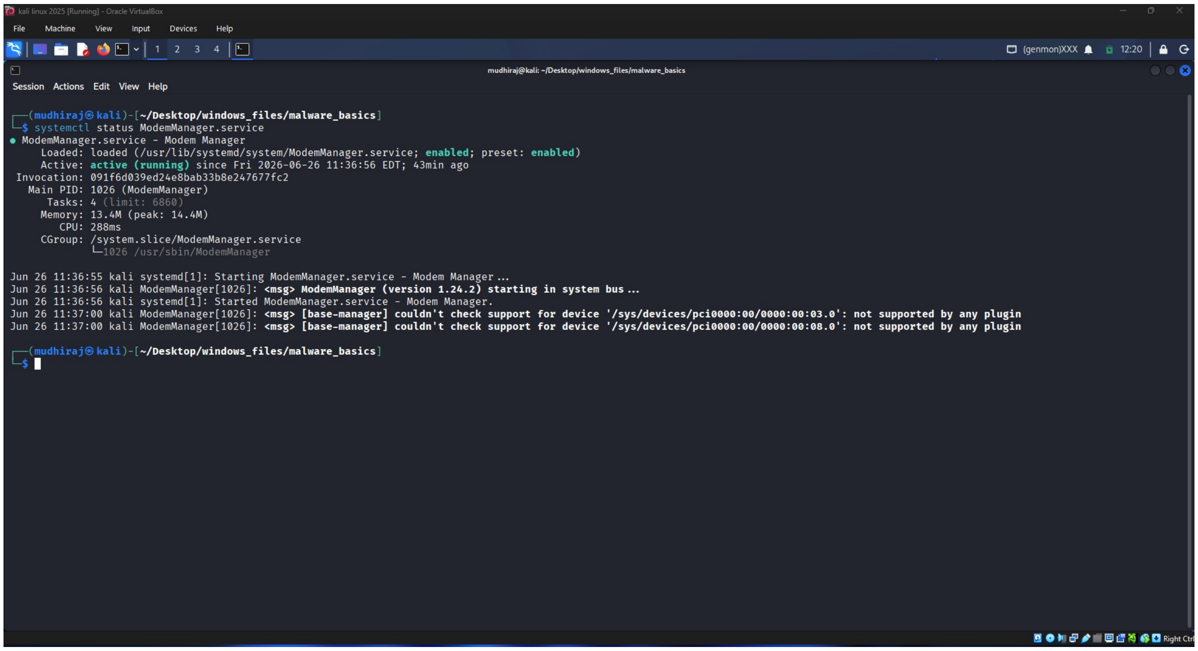
8 – Enabled Services:

Listed enabled system services to identify unused services.



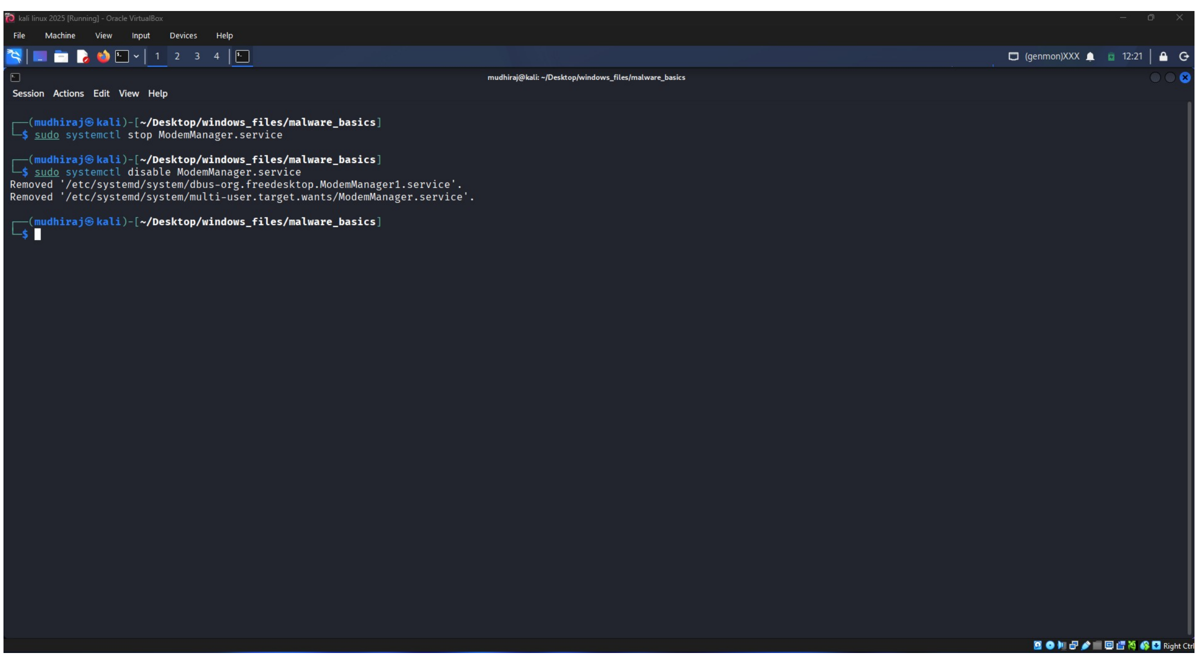
9 – ModemManager Status:

Checked the status of ModemManager before disabling it.



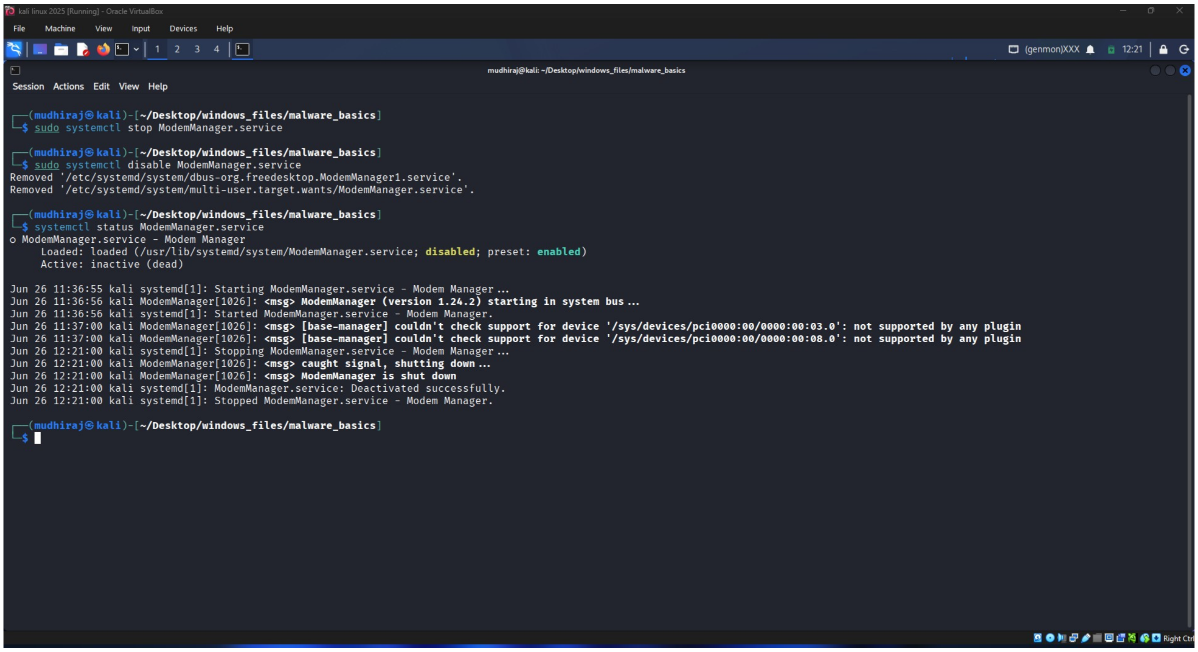
10 – Disable ModemManager:

Stopped and disabled the ModemManager service.



11 – Verification:

Verified ModemManager is disabled and inactive.



Conclusion:

The Kali Linux virtual machine was successfully hardened by applying package updates, cleaning unused packages, configuring basic iptables firewall rules to allow SSH and block Telnet, and disabling the unused ModemManager service. These actions reduce the attack surface and improve the system's security posture.